

AI Career Assistant

Name

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Project Name

AI Career Assistant

Problem Statement

Many job seekers struggle to understand why their resumes are rejected by Applicant Tracking Systems (ATS). They often do not know which skills are missing, how to improve their resumes, or what interview questions they should prepare for. Existing solutions are either expensive or provide limited feedback.

Solution

AI Career Assistant is an intelligent resume analysis platform that uses Google Gemini AI to evaluate resumes and provide personalized career guidance.

Users simply upload their PDF resume, choose their work field, target company, and desired job role. The application analyzes the resume and generates detailed AI-powered insights, including resume evaluation, strengths, weaknesses, missing skills, interview questions, and a personalized learning roadmap.

Core Features

- PDF Resume Upload
 - AI Resume Evaluation
 - AI Evaluation Score
 - Skill Gap Analysis
 - Company-Specific Resume Review
 - Work Field Selection
 - Interview Question Generation
 - Personalized Learning Roadmap
 - Resume Summary
 - Overall AI Feedback
 - Dark & Light Mode
 - Responsive User Interface
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Technology Stack

Frontend

- React.js
- Tailwind CSS
- JavaScript

Backend

- Node.js
- Express.js
- Multer
- PDF Parser

AI

- Google Gemini API
 - Google AI Studio
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Google AI Usage

Google Gemini AI is the core intelligence behind the application.

The application sends extracted resume content along with the selected work field, company, and job role to Gemini.

Gemini performs:

- Resume understanding
- Context-aware evaluation
- Strength detection
- Weakness identification
- Missing skill analysis
- Interview question generation
- Personalized learning roadmap generation
- Overall career feedback

Without Gemini, the application cannot perform intelligent resume analysis.

Workflow

1. User uploads a PDF resume.
2. Backend extracts text from the PDF.
3. User selects:

4. Work Field
 5. Analysis Type
 6. Company (optional)
 7. Job Role
 8. Resume content and user selections are sent to Google Gemini.
 9. Gemini analyzes the resume.
 10. Structured JSON is returned.
 11. Dashboard displays AI-generated results.
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Architecture

User



React Frontend



Express Backend



PDF Text Extraction



Google Gemini API



Structured JSON Response



Interactive Dashboard

Innovation

Unlike traditional resume checkers, AI Career Assistant provides contextual analysis based on work field, company, and job role.

It not only evaluates resumes but also prepares users for interviews by generating customized interview questions and a personalized learning roadmap.

Future Enhancements

- Resume Builder
- Job Description Matching
- ATS Rule Engine
- Resume Version Comparison
- Cover Letter Generator
- LinkedIn Profile Analysis
- AI Career Chatbot
- Resume Score History
- Multi-language Support

Challenges Faced

- Reliable PDF text extraction
- AI response consistency
- Handling API retries
- Structured JSON parsing
- Responsive UI design
- Dark/Light mode support
- Optimizing prompt engineering for accurate analysis

Impact

AI Career Assistant helps students, freshers, and professionals improve their resumes, identify missing skills, prepare for interviews, and receive personalized career guidance using AI.

Conclusion

AI Career Assistant demonstrates how Google Gemini AI can simplify career preparation by transforming a static resume into an intelligent career analysis platform. It combines AI-powered evaluation, personalized recommendations, and interview preparation into a single easy-to-use application.